

AN ARDUINO BASED BRAILLE DISPLAY FOR THE VISUALLY IMPAIRED

A Project report submitted in partial fulfilment of the requirement for the degree of the

BACHELOR OF TECHNOLOGY

IN

ELECTRONICS AND COMMUNICATION ENGINEERING

SUBMITTED BY

SARAGADAM PRUDHVI TEJA (Regd. No.: 319106512045)

CHAGANTIPATI TARUN (Regd. No.: 319106512007)

SANJAY RAMASWAMY GOTHIREDDY (Regd. No.: 319106512044)

Under the esteemed guidance of

Prof. P.V. SRIDEVI, M.E., Ph.D.



**DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING,
ANDHRA UNIVERSITY COLLEGE OF ENGINEERING,
VISA KHAPATNAM-530003**

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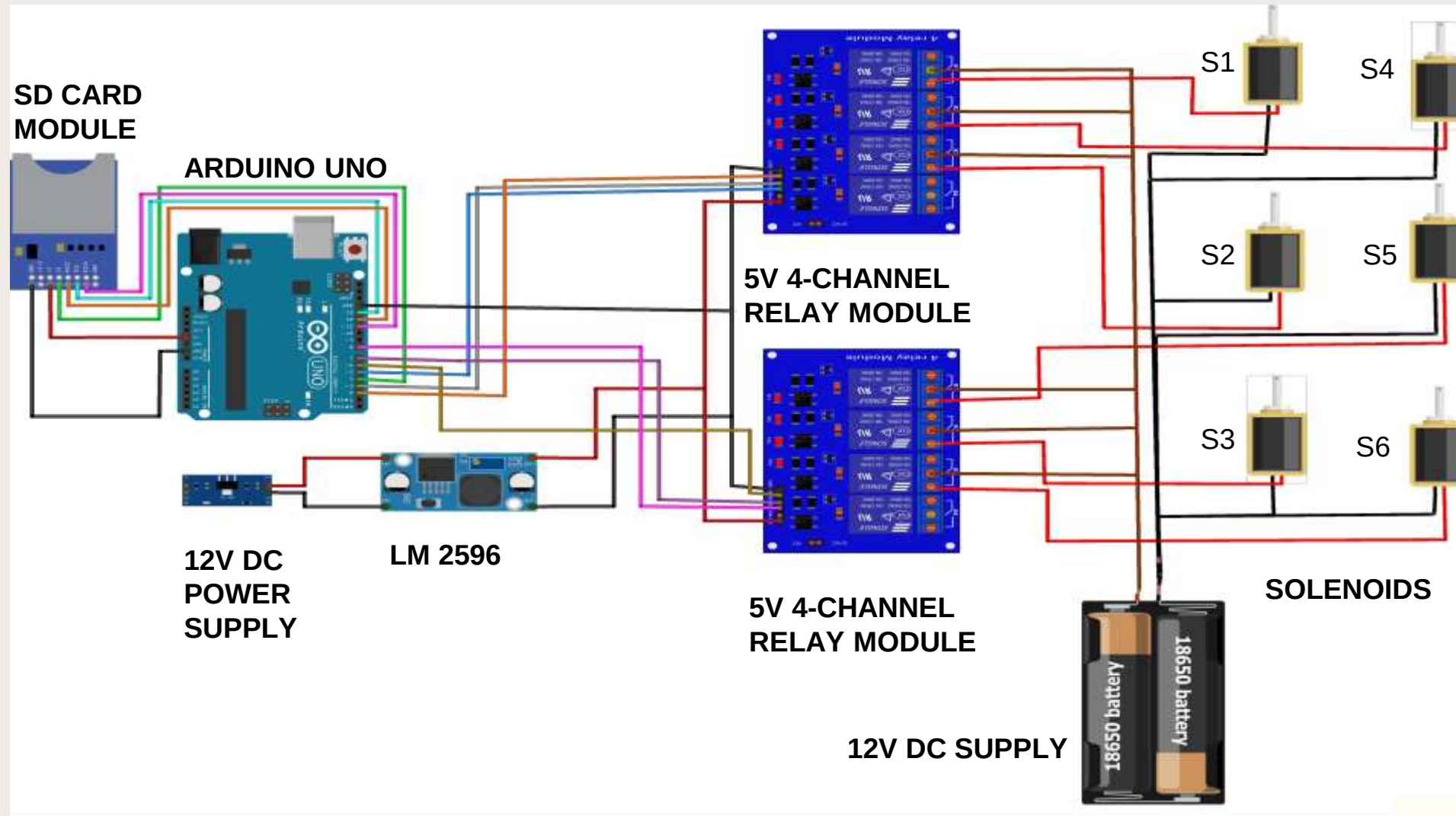
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ABSTRACT

- The main objective is to design and develop an Braille tutor device for visually impaired learners.
- This project uses an Arduino which is responsible for running a translation algorithm to convert stored text into braille script.
- The Arduino sends the signal to the respective relays to perform the switching action.
- The push-pull solenoids which are interfaced with relays pops up and creates a refreshable braille display.

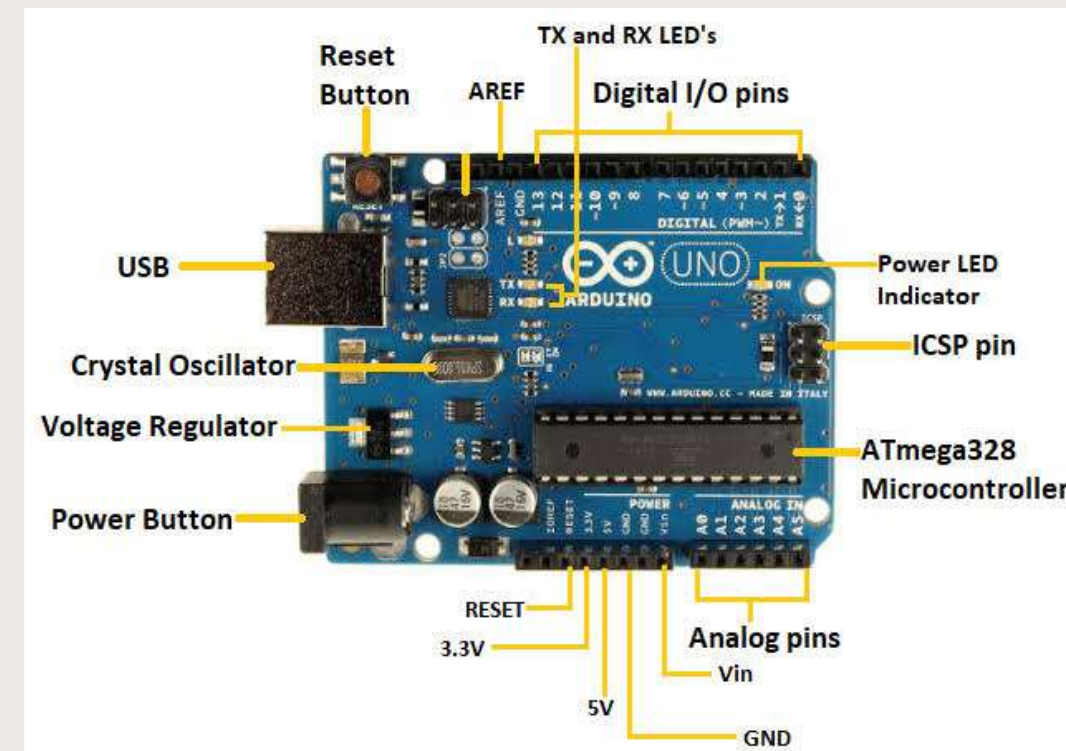
CIRCUIT DIAGRAM OF AN ARDUINO BASED BRAILLE DISPLAY FOR VISUALLY IMPAIRED



COMPONENTS

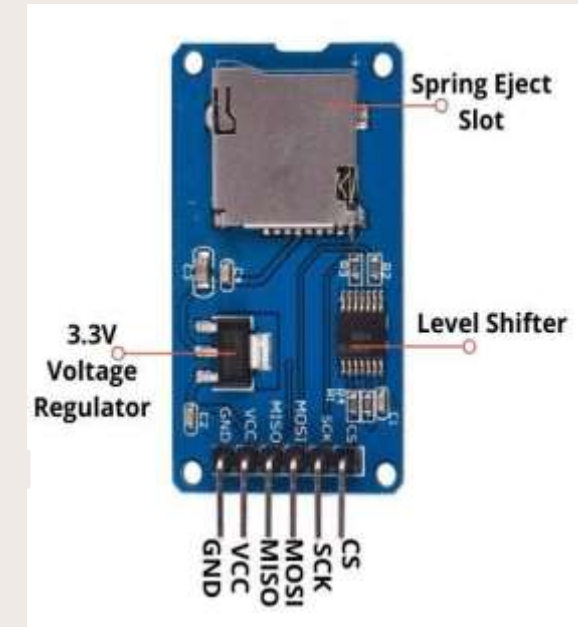
Arduino UNO

- **Arduino** is widely used in a variety of applications due to its simplicity and versatility.
- Arduino UNO is a microcontroller board based on the **ATmega328P**.
- The Arduino IDE is an open-source platform used to write and upload code into the Arduino board.
- The Arduino UNO can be programmed using the C and C++ programming languages, using a standard API which is also known as the Application Programming Interface.



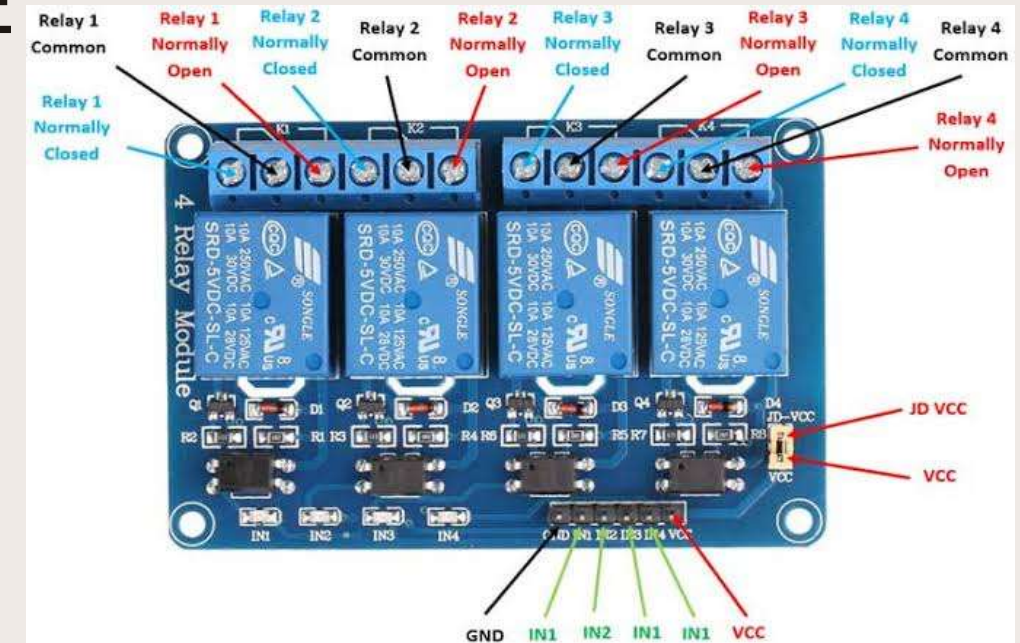
SD-CARD MODULE

- SD Card Module is a breakout board used for SD card processes such as reading and writing with a microcontroller.
- The SD card module is specially useful for projects that require data logging. The Arduino can create a file in an SD card to write and save data using the SD library.
- The SD card module uses the SPI communication protocol.



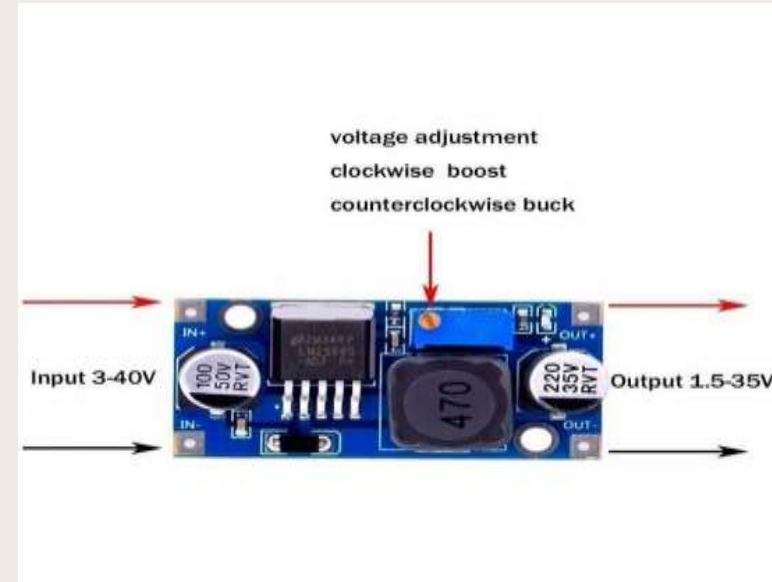
5V 4-CHANNEL RELAY MODULE

- A power relay module is an electrical switch that is operated by an electromagnet.
- The electromagnet is activated by a separate low-power signal from a micro controller. When activated, the electromagnet pulls to either open or close an electrical circuit.



LM2596 BUCK CONVERTER

The LM2596 series of regulators are monolithic integrated circuits that provide all the active functions for a step-down (buck) switching regulator, capable of driving a 3-A load with excellent line and load regulation.



PUSH-PULL SOLENOIDS

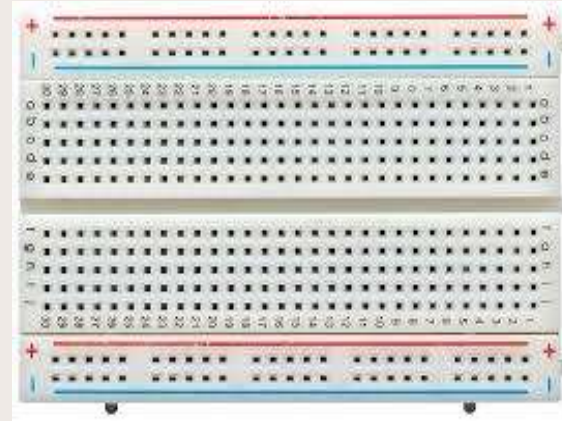
- Push-pull solenoids are on/off type actuators that develop force in one direction when energized.
- The return force must be provided externally, by a return spring for example.



OTHER COMPONENTS



ARDUINO USB
CABLE



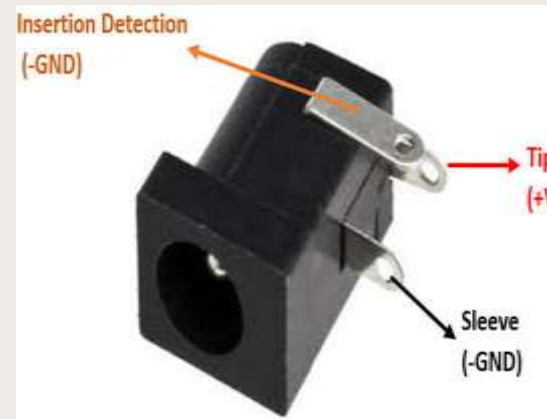
BREAD BOARD



SD CARD



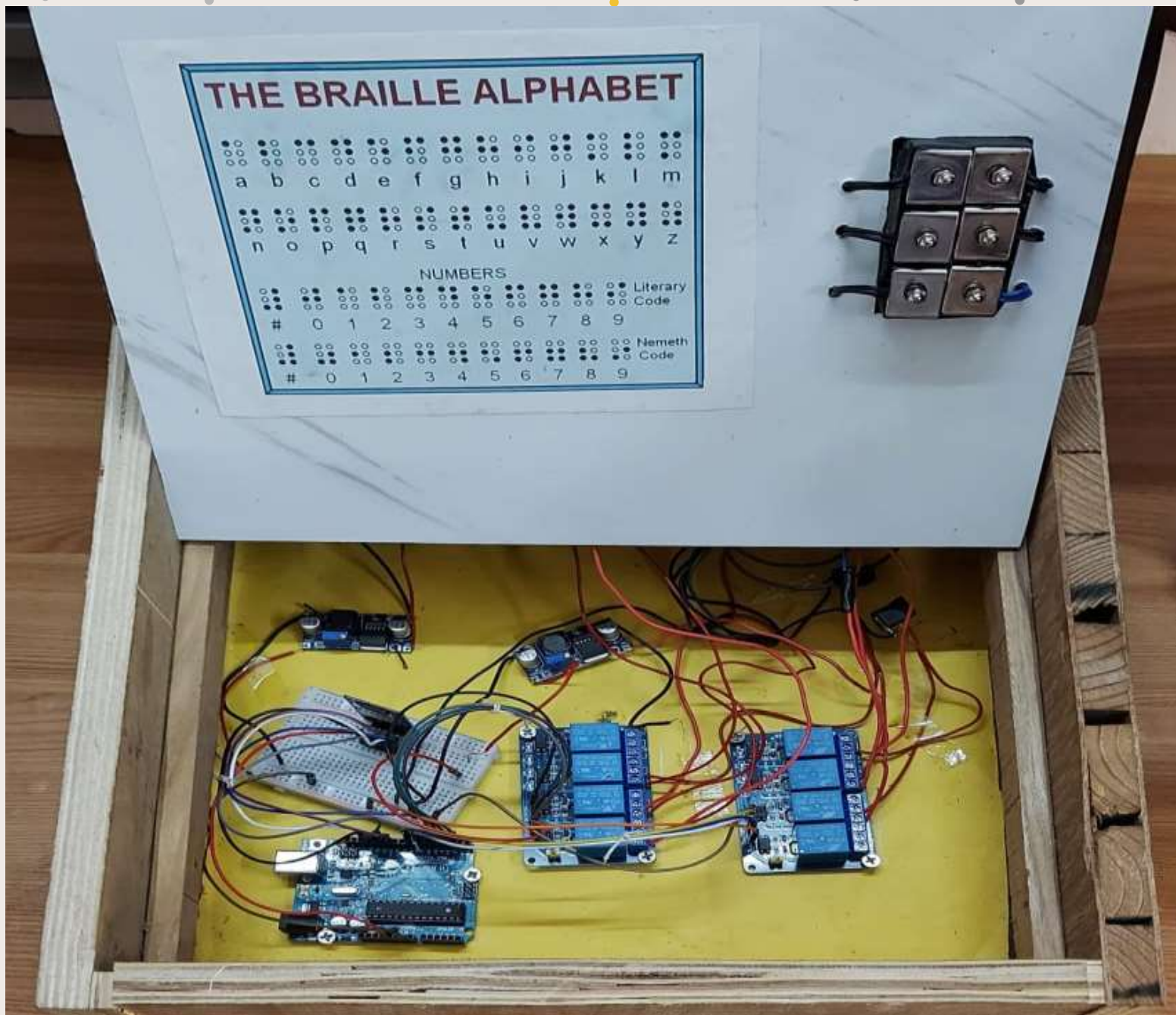
JUMPER WIRES



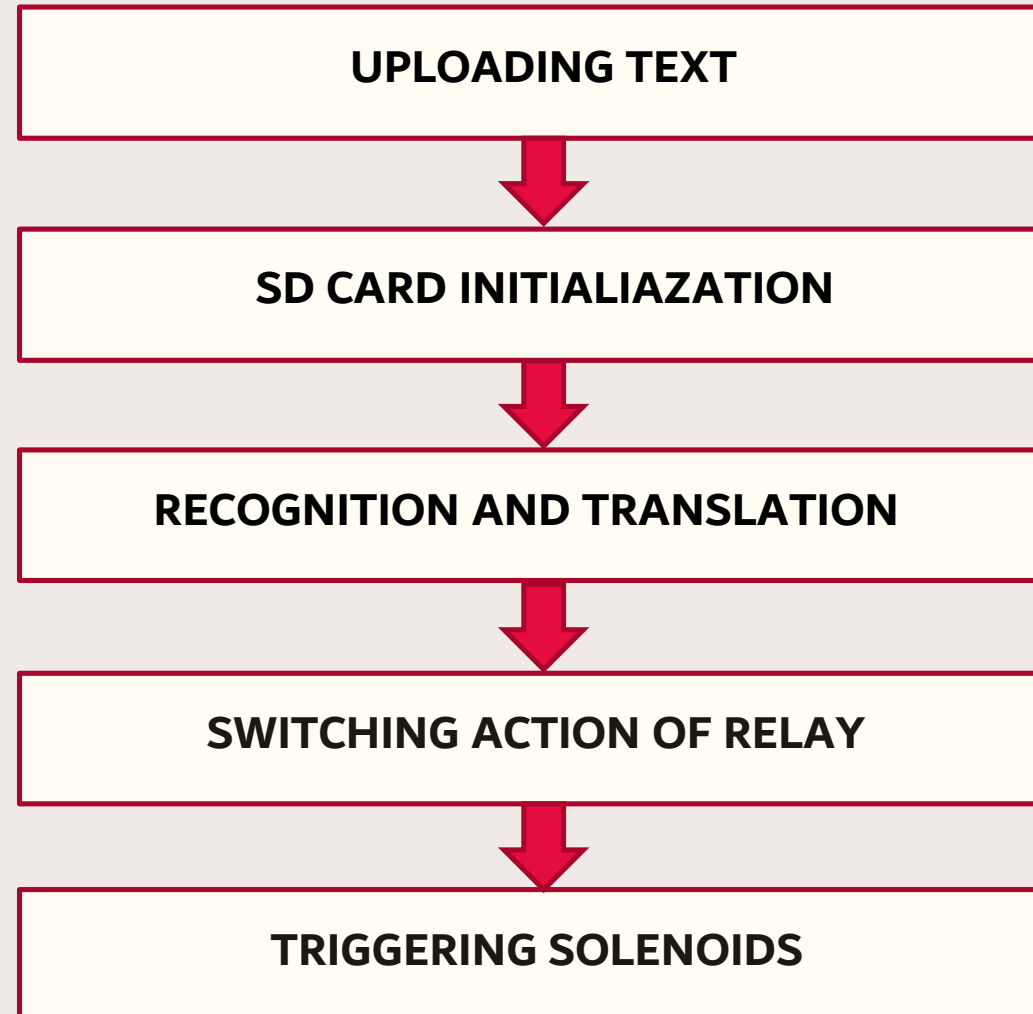
BARREL CONNECTOR



ADAPTOR

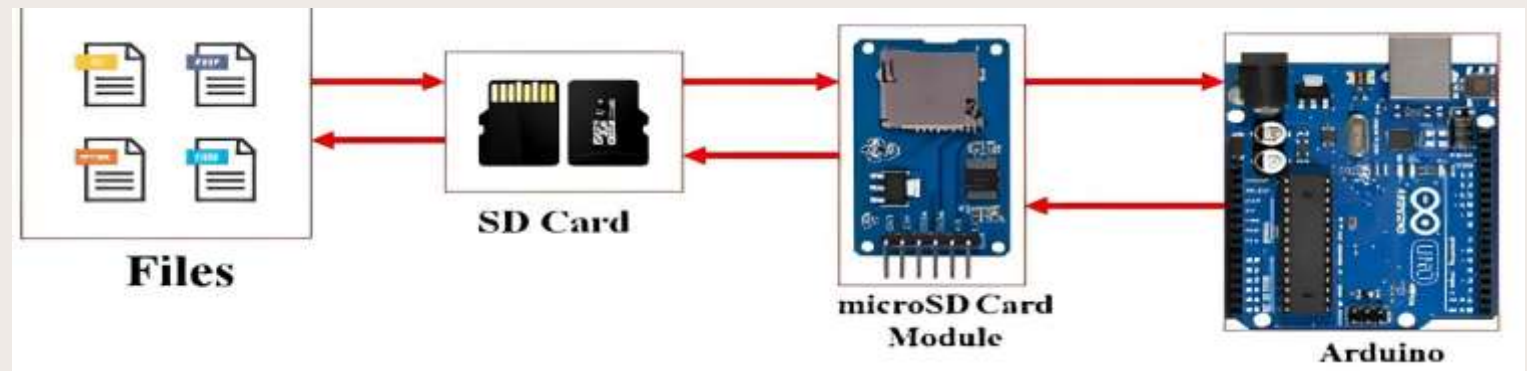


FLOW CHART



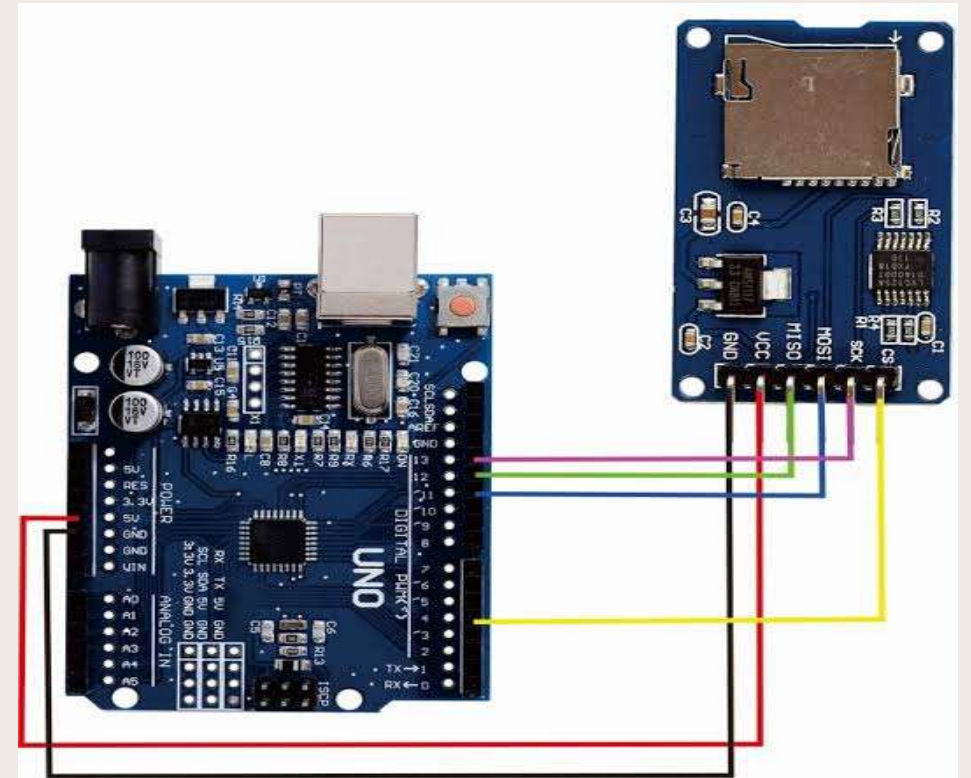
1. UPLOADING TEXT

- In this initial step, the text which should be implemented is saved as a file in '.txt' format since the Arduino UNO can only read '.txt' files.
- This file is then uploaded into SD card via any data transfer device like a simple card reader.
- This SD card is then inserted into the SD card module.
- Here the SD card should be formatted as FAT16 or FAT32 as here SD card module is connected with Arduino.



2. SD-CARD INITIALIZATION

- Here the SD card module and Arduino are connected by SPI pins.
- To enable them 'SPI.h' header file is used.
- The functions provide by the 'SPI.h' library allows the Arduino to send commands to the SD card and manage the flow of data between the two devices.
- Here the chip select wire place a major role for initialization of SD card. If CS pin is connected as defined in the code, then initialization of the SD card is done.



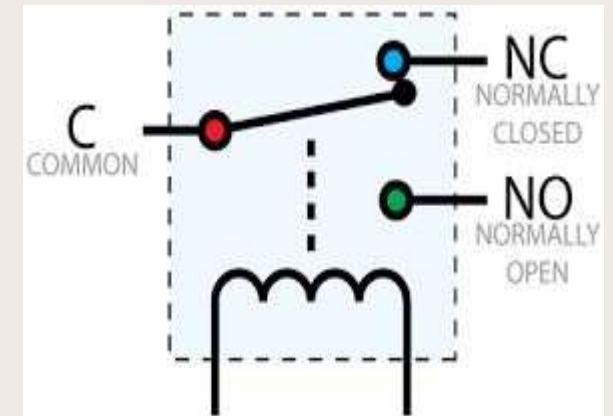


3. RECOGNITION AND TRANSLATION

- After initialisation of the SD card is done, to access SD card information one must use 'SD.h' header file.
- It provides functions for working with SD (Secure Digital) cards on Arduino boards.
- If the file provided in code is available in the SD card, then the corresponding file is opened.
- The Arduino reads character by character and the text in the file can be printed in the serial monitor.
- After the recognition of text in the SD card, each letter in the text is translated into braille script by a user defined function 'displayBraille()'.

4. SWITCHING ACTION OF RELAY

- A relay module is an electrical switch that is operated by an electromagnet.
- The electromagnet is activated by a separate low-power signal from a micro controller which is Arduino.
- When activated, the electromagnet pulls to either open or close an electrical circuit namely relay performs switching action.
- “digitalWrite” in the IDE Coding allows a particular pin to go to HIGH state, in fact allowing it to perform proper switching of Relay.
- In this project ,usually load is connected and operated at NO (Normally Open) Terminal end, so that whenever it goes HIGH, the switching takes place from NC to NO end.

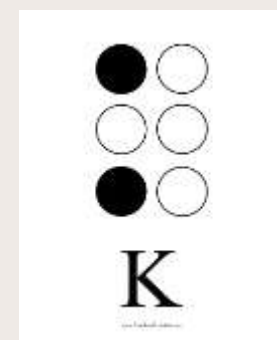
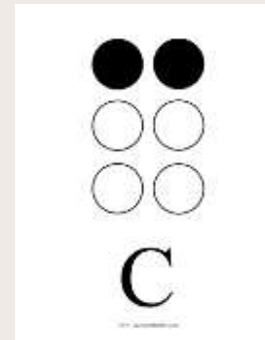
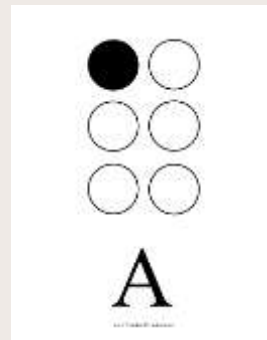
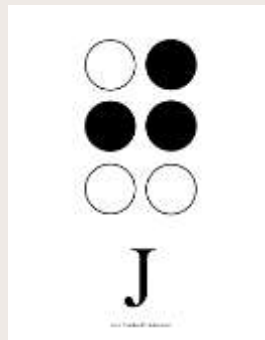


5. TRIGGERING SOLENOIDS

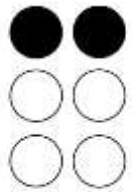
- The push-pull solenoids are connected to NO terminal of relay module.
- The positive terminal of 12V power supply for the solenoids to drive is given to the common terminals of each channel of relay and also all the ground terminals of solenoids are commonly grounded along with negative terminal of 12V power supply and ground pins of both Relay modules and Arduino.
- Based on the switching action of the relay module corresponding solenoids are triggered to create Braille impression for each character.



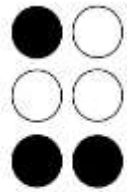
OUTPUT



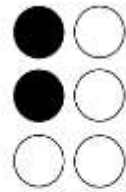
OUTPUT



C



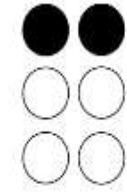
U



B



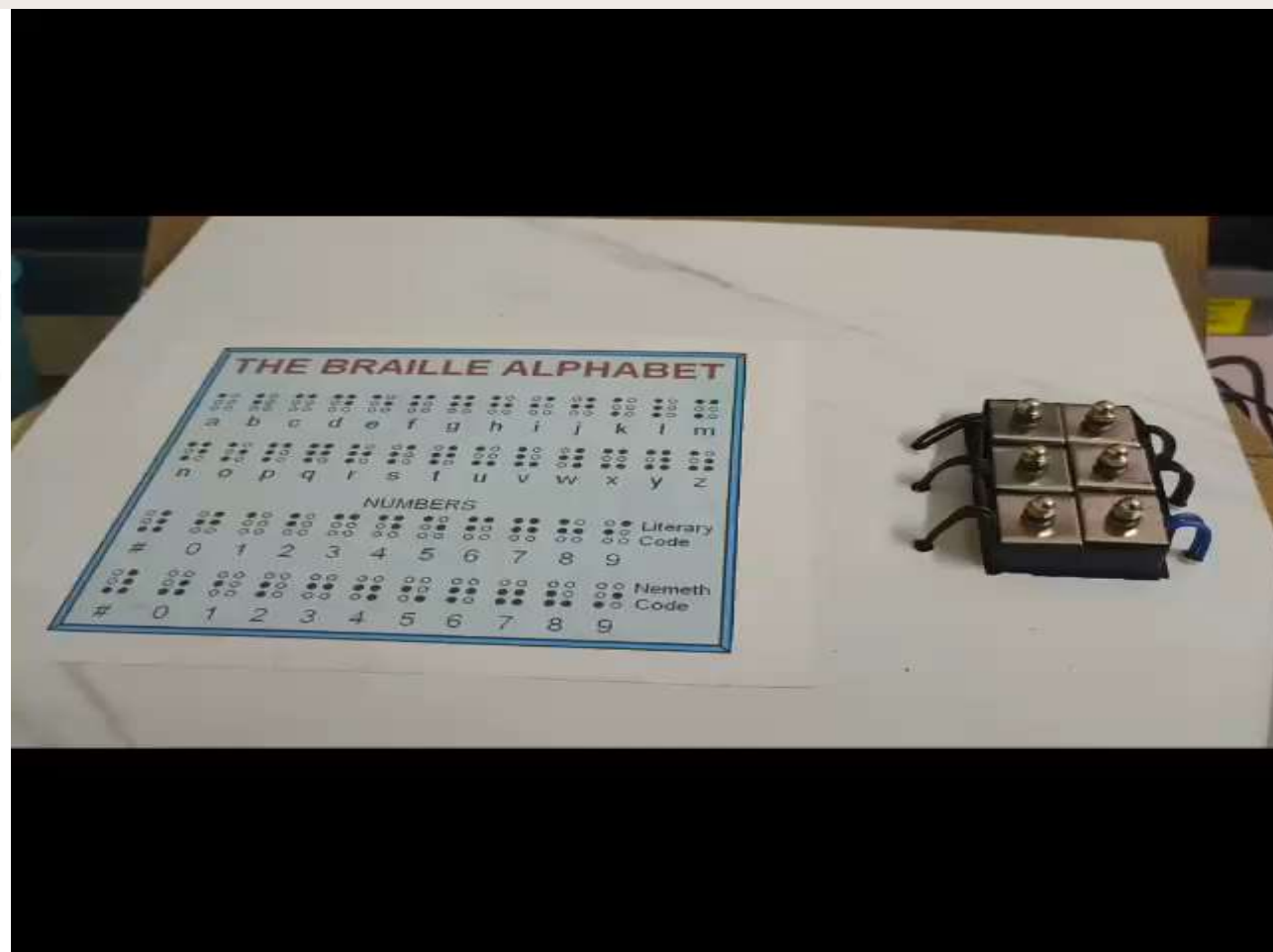
I



C

OUTPUT

A	B	C	D	E	F	G	H	I	J
K	L	M	N	O	P	Q	R	S	T
U	V	W	X	Y	Z	1	2	3	4
5	6	7	8	9	0	.	,	#	space





FUTURE SCOPE

- Adding Bluetooth connectivity would allow users to connect their smartphones to the device and input text using their mobile devices.
- A language translation software can be incorporated into the device to automatically translate text into English and then into braille for output.

CONCLUSION

- In this project the Arduino is interfaced to SD card module to access the text and also interfaced to relays to perform switching action.
- The push-pull solenoids are then interfaced with relay module in order to create refreshable braille impression.
- This technology enables visually impaired people to switch from reading braille books to refreshable braille display.

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THANK YOU

